S2: Distributional representations

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S. Clark. Vector space models of lexical meaning. In S. Lappin and C. Fox, editors, Handbook of Contemporary Semantics — second edition, chapter 16, pages 493–522. Wiley – Blackwell, 2015.

- Lexical semantics, can process information from texts without the need to know the model
- Compare similarity between words
- Origins in linguistics and cognitive science (memory, priming)
- Vector space models and information retrieval
- Basis vector identify dimensions (in comparison to the actual vectors for words that we compare)
- TF/IDF: counter the problem that some words are generally more frequent than others, divide by the number of documents the word occurs in (or multiply by the 1/DF
  - see Karen Spärck Jones
- PMI: another method that measures the strength of association: how often two events co-occur compared to how often they occur independently, see J&M Chapter 6; related to log-likelihood ratio and KL-Divergence, see here
- Normalise the dot product by the length of the documents and hence we get an equation for cosine
  - compare with the tensor product later, see here
  - dot product gives us a single value, see here
- Dimensionality reduction: singular value decomposition = latent semantic analysis/indexing, non-negative matrix factorisation, random indexing: identifying meaningful dimensions, reduces noise, helps with sparsity
- syntagmatic and paradigmatic relations between words, relevance for contexts
- Curran investigates different ways to represent contexts, including POS tags and relations, linguistic processing words such as stemming, removal of stop-words
- We used IDF to weigh dimensions but other collocation statistics can also be used

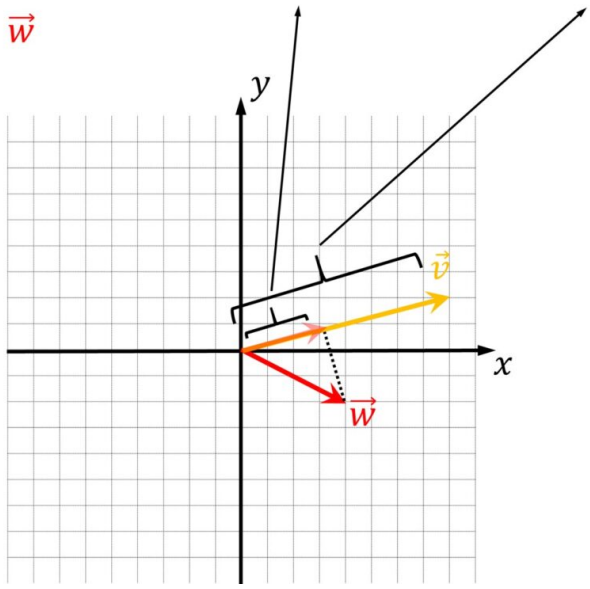
- Normalisation in cosine ignores the frequency effect, it is normalised to the vector length, hence only the angle is important
- The corpus is crucial for the quality of relations
- Does cosine/semantic similarity always identify synonyms?
- Intrinsic and extrinsic evaluation
- Attributions and relational similarity: semantic relatedness vs similarity
- Word sense disambiguation is challenging
- Compositional vector-space models
- But what is a composition of sentence in terms of word contexts?
- Mitchell and Lapata compare various compositional operators, point-wise multiplication works best
- Tensor product comes from Smolensky, it creates larger vector space

$$\mathbf{u} \otimes \mathbf{v} = \mathbf{A} = \begin{bmatrix} u_1 v_1 & u_1 v_2 & \dots & u_1 v_n \\ u_2 v_1 & u_2 v_2 & \dots & u_2 v_n \\ \vdots & \vdots & \ddots & \vdots \\ u_m v_1 & u_m v_2 & \dots & u_m v_n \end{bmatrix}$$

$$\vec{v} \otimes \vec{w} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 4 & 5 \end{bmatrix} = \begin{bmatrix} 1 \cdot 4 & 1 \cdot 5 \\ 2 \cdot 4 & 2 \cdot 5 \\ 3 \cdot 4 & 3 \cdot 5 \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ 8 & 10 \\ 12 & 15 \end{bmatrix}$$

- Dot product

$$\underset{\vec{v}}{\begin{bmatrix}8\\2\end{bmatrix}} \cdot \underset{\vec{w}}{\begin{bmatrix}4\\-2\end{bmatrix}} = (\text{Length of projected } \vec{w}) \cdot (\text{Length of } \vec{v})$$



$$\begin{bmatrix}2\\7\\1\end{bmatrix} \cdot \begin{bmatrix}8\\2\\8\end{bmatrix} = 2 \cdot 8 + 7 \cdot 2 + 1 \cdot 8 = 38$$

Dot product

- point-wise matrix multiplication is multiplying each individual position of two matrices to create another matrix of the same dimension as the source matrices where the positions are products. (This is different than matrix multiplication).
- Hence, we have 3 different ways to combine information; we reduce dimensions, increase dimensions or keep the same dimensions. This will be relevant for modelling language with neural networks.
- Combinatory categorical grammar / CCG / Lambek pre-group grammar
- Grammar is defined as operations over semantic types: what vector representations correspond to the semantic types of the grammar?
- Tensor product is a multiplication of every dimension of the first space with every dimension of the second space, therefore the result is much larger, effectively comparing every dimension with every other dimension
- Verb is relational in nature, it relates a particular subject the particular objects, hence the tensor product expresses these interactions
- Running example with plausibility vectors
- Baroni and Zamparelli, adjectives are derived and nouns are vectors

- As composition is applied the semantic types are becoming smaller, i.e. chasing cat events
- How do we evaluate compositions? Inference

"Even taking account of laudable attempts such as Bos & Markert (2006), building on earlier work such as Alshawi (1992), it is probably fair to say that the classical logic-based enterprise in natural language processing has failed."

Interesting references

- Bruni, E., G. Boleda, M. Baroni, & N. Tran. 2012 (2012), Distributional semantics in technicolor, in Proceedings of the 50th Annual Meeting of the Association for Computational Linguistics, Jeju Island, Korea, (136–145).
- Pulman, Stephen (2012), Distributional semantic models, in Sadrzadeh Heunen & Grefenstette (eds.), Compositional Methods in Physics and Linguistics, Oxford University Press.
- tf-idf, Point-wise mutual information (PMI), SVD
- Chapter 6 Vector semantics and embeddings. D. Jurafsky and J. H. Martin. Speech and language processing: an introduction to natural language processing, computational linguistics, and speech recognition. Third edition draft, Stanford University and University of Colorado at Boulder, December 30 2020.
- M. Baroni. Composition in distributional semantics. Language and Linguistics Compass, 7(10):511–522, 2013.
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